Luigui **Gallardo-Becerra**

BIOINFORMATICIAN | MOLECULAR BIOLOGIST

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Experienced Bioinformatician and Data Scientist with 6+ years of expertise in genomics, transcriptomics, and microbiome analysis. Skilled in developing scalable bioinformatics pipelines, managing large-scale NGS datasets, and implementing machine learning approaches for biological data analysis. Proven track record with 14 peer-reviewed publications and extensive experience in high-performance computing environments.

Experience

Research Assistant (Bioinformatics & Data Science)
Institute of Biotechnology, UNAM
January 2019 - July 2025

Bioinformatics workflow engineering: Designed and implemented automated, reproducible pipelines using Snakemake, Nextflow, Python, and Bash for large-scale NGS data processing.

NGS data analysis: Processed and analyzed whole-genome sequencing, RNA-seq, 16S rRNA, metagenomic, and metatranscriptomic datasets, including QC, alignment, quantification, and downstream interpretation.

Microbiome and metagenomics: Performed taxonomic profiling, functional annotation, and comparative microbiome analyses to identify microbial signatures and biological patterns.

Machine learning and statistical modeling: Applied supervised and unsupervised learning methods to identify biomarkers, classify samples, and uncover structure in high-dimensional omics datasets.

Database design and curation: Built and maintained biological databases; developed ETL workflows for structured multi-omics data integration; ensured data consistency and quality control.

High-performance computing (HPC): Administered HPC clusters, optimized pipeline performance, and managed large-scale computational workloads across Linux/SLURM environments.

Scientific software development: Created custom tools and analysis scripts in Python, R, and Perl to address project-specific computational challenges and automate complex workflows.

Data Engineer
Appen
October 2016 - December 2018

Supported large-scale data processing projects by developing automated ETL pipelines to ensure efficient integration, cleaning, and transformation of high-volume datasets.

Implemented data quality control workflows used for machine-learning model development, improving reliability and reproducibility of downstream analyses.

Designed automated data collection and validation systems that reduced manual workload, increased data accuracy, and streamlined reporting processes across projects.

Software Engineer (Internship)
Dept. of Computer Science, CUCEI
January 2016 - July 2016

Developed a cloud-based web application using ASP.NET Core for managing patient records and associated clinical data, contributing to database design, backend development, and user interface implementation.

Built features to support secure data storage, retrieval, and visualization, reinforcing best practices in data integrity and compliance—skills directly applicable to modern bioinformatics and laboratory information systems.

Education

National Autonomous University of Mexico (UNAM) *Mexico City, Mexico*
PH.D. IN COMPUTATIONAL BIOLOGY *January 2019 - July 2025*

National Autonomous University of Mexico (UNAM) *Mexico City, Mexico*
MASTER OF SCIENCE IN COMPUTATIONAL BIOLOGY *August 2016 - January 2019*

University of Guadalajara (UDG) *Guadalajara, Mexico*
BACHELOR OF SCIENCE IN MOLECULAR BIOLOGY *August 2012 - January 2016*

Key Skills

Programming & Scripting:
Python, R, Bash, SQL, Perl, C++, Java, JavaScript, HTML/CSS

Bioinformatics Tools & Frameworks:
Snakemake, Nextflow, Galaxy, QIIME2, Bioconductor, Biopython, Samtools, BLAST, BWA, Bowtie2

NGS Data Analysis:
Illumina/PacBio workflows; quality control (FastQC, MultiQC); read alignment; variant calling; de novo assembly; genome/transcriptome annotation

Microbiome & Multi-Omics Analysis:
Metagenomics, metatranscriptomics, 16S rRNA profiling, comparative genomics, phylogenetics, transcriptomics

Data Engineering & Management:
SQL/NoSQL database design, ETL workflows, data warehousing, Git version control, reproducible research practices

Computing & Infrastructure:
Linux/Unix, HPC cluster administration, cloud computing (AWS, Google Cloud), containerization (Docker, Singularity), Conda environments

Data Science & Visualization:
Statistical modeling, machine learning, visualization with ggplot2, seaborn, matplotlib, Tableau, and R Shiny

Professional Skills:
Project management, scientific communication, cross-functional collaboration, technical writing, problem-solving

Languages:
English (Full professional proficiency), Spanish (Native)

Selected Publications (14 total)

Gallardo-Becerra L, Cornejo-Granados F, Bikel S, Arenas I, López-Leal G, Alvarado-Gonzalez C, et al. Bioactive plasmid- and phage-encoded antimicrobial peptides (AMPs) in the human gut: a metatranscriptome–virome profiling reveals exploratory links to metabolic human diseases. Microb Ecol. 2025 Nov 28.

Gallardo-Becerra L, Cornejo-Granados F, Romero-Hidalgo S, Lopez-Zavala AA, Cota-Huízar A, Cervantes-Echeverría M, et al. Host genome drives the microbiota enrichment of beneficial microbes in shrimp: exploring the hologenome perspective. Anim Microbiome. 2025 May 22;7(1):50.

Gallardo-Becerra L, Cervantes-Echeverría M, Cornejo-Granados F, Vazquez-Morado LE, Ochoa-Leyva A. Perspectives in searching antimicrobial peptides (AMPs) produced by the microbiota. Microb Ecol. 2023 Dec 1;87(1):8.

Gallardo-Becerra L, Cornejo-Granados F, García-López R, Valdez-Lara A, Bikel S, Canizales-Quinteros S, et al. Metatranscriptomic analysis to define the Secrebiome, and 16S rRNA profiling of the gut microbiome in obesity and metabolic syndrome of Mexican children. Microb Cell Fact. 2020 Dec;19(1):61.

Gallardo-Becerra L, Cornejo-Granados F, Leonardo-Reza M, Ochoa-Romo JP, Ochoa-Leyva A. A meta-analysis reveals the environmental and host factors shaping the structure and function of the shrimp microbiota. PeerJ. 2018 Aug 10;6:e5382.